# Computer Vision

From pixels and filters to detection, segmentation, OCR, video and vision-language models.

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# 1. Images as arrays: pixels, channels, colour spaces [Beginner]

### Explanation (English)
An image is an H x W x C array. Colour spaces: RGB, BGR (OpenCV), HSV, grayscale.

### Tanglish (Tamil + English) Vilakkam
Image = pixel array. OpenCV BGR, matplotlib RGB - mix aaga koodadhu.

### Example / Program
```
import cv2,numpy as np
img=np.zeros((100,100,3),np.uint8);img[:,:,2]=255
print(img.shape,cv2.cvtColor(img,cv2.COLOR_BGR2HSV)[0,0])
```

### Learn more
YouTube search: Images as arrays: pixels, channels, colour spaces tutorial

# 2. Image processing: resize, crop, blur, threshold, morphology [Beginner]

### Explanation (English)
Gaussian/median blur, binary & adaptive thresholding, erosion/dilation, histogram equalisation.

### Tanglish (Tamil + English) Vilakkam
Resize, blur, threshold - preprocessing basics.

### Example / Program
```
import cv2,numpy as np
g=(np.random.rand(50,50)*255).astype('uint8')
b=cv2.GaussianBlur(g,(5,5),0);_,t=cv2.threshold(b,127,255,cv2.THRESH_BINARY)
```

### Learn more
YouTube search: Image processing: resize, crop, blur, threshold, morphology tutorial

# 3. Edges, contours, Hough transform, feature descriptors (SIFT/ORB) [Intermediate]

### Explanation (English)
Canny edges, findContours, Hough lines/circles, keypoints and descriptors for matching and stitching.

### Tanglish (Tamil + English) Vilakkam
Edge, contour, SIFT/ORB features.

### Example / Program
```
import cv2
orb=cv2.ORB_create()
# kp,des=orb.detectAndCompute(gray,None)
```

### Learn more
YouTube search: Edges, contours, Hough transform, feature descriptors (SIFT/ORB) tutorial

# 4. Image classification with CNNs [Beginner]

### Explanation (English)
Train a CNN or fine-tune a pretrained backbone (ResNet, EfficientNet, ViT) on labelled images.

### Tanglish (Tamil + English) Vilakkam
Image classification: CNN / transfer learning.

### Example / Program
```
import torchvision.transforms as T
tf=T.Compose([T.Resize(224),T.ToTensor(),T.Normalize([.485,.456,.406],[.229,.224,.225])])
```

### Learn more
YouTube search: Image classification with CNNs tutorial

## 5. Data augmentation [Beginner]

### Explanation (English)

Flips, crops, rotations, colour jitter, MixUp, CutMix, RandAugment (Albumentations) improve generalisation.

### Tanglish (Tamil + English) Vilakkam

Augmentation data increase panni overfitting kuraikkum.

### Example / Program

```
import albumentations as A
aug=A.Compose([A.HorizontalFlip(),A.RandomBrightnessContrast(),A.Rotate(15)])
```

### Learn more

YouTube search: Data augmentation tutorial

## 6. Object detection: R-CNN family, SSD, YOLO, DETR [Intermediate]

### Explanation (English)

Predict bounding boxes + classes. Two-stage (Faster R-CNN) vs one-stage (YOLO, SSD) vs transformer (DETR). Metrics: IoU, mAP, NMS.

### Tanglish (Tamil + English) Vilakkam

Object detection: box + class. YOLO real-time.

### Example / Program

```
from ultralytics import YOLO
m=YOLO('yolov8n.pt')
r=m('https://ultralytics.com/images/bus.jpg');print(r[0].boxes.cls)
```

### Learn more

YouTube search: Object detection: R-CNN family, SSD, YOLO, DETR tutorial

## 7. Semantic, instance and panoptic segmentation (U-Net, Mask R-CNN, SAM) [Advanced]

### Explanation (English)

Pixel-level labelling. U-Net for medical imaging; Mask R-CNN for instances; Segment Anything (SAM) for promptable segmentation.

### Tanglish (Tamil + English) Vilakkam

Segmentation: ovvoru pixel-um classify. U-Net, SAM.

### Learn more

YouTube search: unet segmentation pytorch tutorial

## 8. Face detection and recognition [Intermediate]

### Explanation (English)

Detect faces (Haar, MTCNN, RetinaFace) and recognise via embeddings (FaceNet, ArcFace). Consider privacy/bias.

### Tanglish (Tamil + English) Vilakkam

Face detect + embeddings. Privacy care pannunga.

### Example / Program

```
import cv2
c=cv2.CascadeClassifier(cv2.data.haarcascades+'haarcascade_frontalface_default.xml')
```

### Learn more

YouTube search: Face detection and recognition tutorial

## 9. OCR and document AI [Intermediate]

### Explanation (English)

Extract text with Tesseract, EasyOCR, PaddleOCR, TrOCR; layout models (LayoutLM, Donut) for invoices and forms.

### Tanglish (Tamil + English) Vilakkam

Image-la irundhu text edukka OCR.

### Example / Program

```
import pytesseract
from PIL import Image
# print(pytesseract.image_to_string(Image.open('a.png')))
```

### Learn more

YouTube search: OCR and document AI tutorial

## 10. Pose estimation, tracking, optical flow, video understanding [Advanced]

### Explanation (English)

MediaPipe/OpenPose keypoints; trackers (SORT, DeepSORT, ByteTrack); optical flow; video transformers.

### Tanglish (Tamil + English) Vilakkam

Pose, tracking, video analysis.

### Example / Program

```
import mediapipe as mp
hands=mp.solutions.hands.Hands()
```

### Learn more

YouTube search: Pose estimation, tracking, optical flow, video understanding tutorial

## 11. Vision Transformers (ViT), CLIP and vision-language models [Advanced]

### Explanation (English)

ViT splits images into patches and applies a Transformer. CLIP aligns images and text for zero-shot classification; VLMs (LLaVA, GPT-4V style) answer questions about images.

### Tanglish (Tamil + English) Vilakkam

ViT patches use pannum. CLIP image-text match.

### Example / Program

```
from transformers import pipeline
z=pipeline('zero-shot-image-classification')
# z('a.jpg',candidate_labels=['cat','dog'])
```

### Learn more

YouTube search: Vision Transformers (ViT), CLIP and vision-language models tutorial

## 12. 3D vision, depth, SLAM, point clouds, NeRF [Advanced]

### Explanation (English)

Stereo/monocular depth, SLAM for robots, PointNet, NeRF and Gaussian splatting for 3D reconstruction.

### Tanglish (Tamil + English) Vilakkam

3D, depth, SLAM, NeRF - advanced.

### Learn more

YouTube search: nerf explained 3d reconstruction

## 13. Image generation and editing (GAN, diffusion, inpainting, ControlNet) [Advanced]

**Explanation (English)**
Text-to-image, inpainting, super-resolution, style transfer.

**Tanglish (Tamil + English) Vilakkam**
Image generation: diffusion, inpainting, super-resolution.

**Learn more**
YouTube search: stable diffusion diffusers tutorial

## 14. Medical, satellite, industrial and autonomous-driving vision [Advanced]

**Explanation (English)**
Domain challenges: scarce labels, class imbalance, 3D/volumetric data, strict safety requirements.

**Tanglish (Tamil + English) Vilakkam**
Domain-specific vision: medical, satellite, driving.

**Learn more**
YouTube search: computer vision applications medical satellite

## 15. Deploying vision models: ONNX, TensorRT, TFLite, edge devices [Advanced]

**Explanation (English)**
Convert and optimise for Jetson, mobile, browsers (TF.js / ONNX Web).

**Tanglish (Tamil + English) Vilakkam**
Edge devices-la deploy: TFLite, ONNX, TensorRT.

**Example / Program**

```
# model.export(format='onnx')  # ultralytics
```

**Learn more**
YouTube search: Deploying vision models: ONNX, TensorRT, TFLite, edge devices tutorial